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Commentary

The good, the bad, and the ugly of consumer sleep technologies use among athletes: A call for action

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1. Introduction

For athletes, the quality and quantity of sleep are essential elements for optimizing recovery and subsequent performance.[1] Unfortunately, athletes frequently face challenges getting sufficient, high-quality sleep.[2] A recent systematic review and meta-analysis of 81 studies involving 1830 athletes concluded that compared to healthy non-athletes, athletes' sleep duration was shorter with poorer sleep efficiency.[3] Notable sleep issues (e.g., sleep apnea and snoring, as well as insomnia and trouble falling asleep) were revealed in young athletes.[3] It was common for sleep quality and architecture to change over time depending on the training period.[3]

2. The good

Understanding how and to what extent sleep can be enhanced in order to improve athletic performance requires the ability to accurately measure and quantify it.[1] Polysomnography (PSG) is the gold standard for the assessment of sleep/wake patterns.[1] However, it is expensive, cumbersome, and requires a sleep specialist, making it impractical for regular use by athletes. Luckily, the last decade has seen an increasing proliferation of publicly available consumer sleep technologies (CSTs), which includes sleep recording devices that can be categorized as wearables and non-wearables (also known as "nearables").[4]

Wearable devices are worn on the athlete's wrist (i.e., a watch), finger (i.e., a ring), head (e.g., a headband), or chest (i.e., a belt).[4] Actigraphy, which combines activity monitors and sleep/wake diaries, is the most widely used wearable alternative to PSG. Accelerometers are incorporated into this device type to determine periods of "movement" (i.e., awake) and/or "nonmovement" (i.e., asleep).[5]

For the new generation of wrist−worn wearables (e.g., commercial smartwatches that are sensor-based and most commonly used in athletic settings[6]), sleep is tracked not only by movement but also by pulse oximetry, skin temperature, heart rate, and heart rate variability (among other physiological measures) using manufacturer algorithms.[4] Contrary to actigraphy-only devices, multi-sensor devices allow for the prediction of sleep stages above and beyond simple binary sleep/wake distinctions.[7]

Nearables are unobtrusive devices that detect sleep from a distance and are typically placed in the bed or under the mattress.[8] Nearables typically estimate sleep stages by sensing movement, breathing, cardiovascular m (e.g., heart rate, oxygen saturation, stroke volume, and even an electrocardiogram records), and sound, as well as ambient light, humidity, and temperature.[9]

3. The bad

The use of CSTs by athletes can often be problematic. First, the accuracy of CSTs for monitoring sleep is questionable. In this context, Chinoy et al.[10] tested the performance of a number of CSTs, including 5 wearables (i.e., Philips Respironics

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Actiwatch 2, Fatigue Science Readiband, Garmin Fenix 5S, Fitbit Alta HR, and Garmin Vivosmart 3) and 3 nearables (i.e., EarlySense Live, SleepScore Max, and ResMed S+) *vs.* PSG as the gold standard sleep assessment technique. The authors found that CSTs had high sensitivity but relatively lower specificity, indicating a tendency for the devices to accurately detect sleep but to detect wakefulness less accurately as compared with PSG.[10] Specifically, an overestimation of total sleep time and sleep efficiency, as well as an underestimation of wake time (i.e., wake after sleep onset) were observed for most CSTs. The ResMed S+ and Fitbit Alta HR devices were the most accurate in terms of estimating total sleep time.[10] Additionally, significant differences were observed in all major summary metrics compared to those obtained using actigraphy and PSG.[10] While many CSTs did not significantly differ from PSG, their observed differences were even smaller when compared to actigraphy.[10] It should also be noted that the Garmin devices had more extreme measurement differences compared to objective clinical measures of sleep (i.e., PSG or actigraphy);[10] therefore, using these particular CSTs for assessing the sleep of athletes should be done with caution.

4. The ugly

In addition to the possible inaccurate sleep metrics provided by some CSTs, a perfectionist athlete with personality traits characterized by excessively high personal standards and extreme critical self-evaluation[11] will very likely endeavor to pursue optimal sleep metrics based on their use of CSTs. This obsessive pursuit of ideal sleep has recently been termed "orthosomnia,"[12,13] a combination of "ortho" referring to straight, proper, or right, and "somnia" referring to sleep.[13] Investigations of orthosomnia are in their infancy. Nevertheless, an obsessional preoccupation with sleep tracker data, frequent checking of the sleep tracker, and concern and anxiety about separation from one's CST are examples of behaviors that may commonly be associated with orthosomnia.[12] An athlete with orthosomnia is continuously obliged to use a laptop, smartphone, or tablet to collect and visualize their sleep. However, such behaviors may lead to and/or exacerbate an emerging related psychological condition: the fear of lack of access to a mobile/smartphone and or internet connectivity, which is known as nomophobia.[14] In a recent systematic review and meta-analysis by our research group, we found that approximately 20% of people had mild symptoms of nomophobia, 50% had moderate symptoms, and 20% had severe symptoms.[14] Despite the lack of studies investigating the prevalence of nomophobia in athletes, the use of fitness/sleep trackers often connected to a mobile phone suggests that a significant percentage of athletes may suffer from severe nomophobia. Previous studies by our research team showed a strong association between nomophobia, anxiety, and insomnia.[15,16]

Insomnia symptoms are reported by up to 70% of professional athletes and are common in both individual and team sports.[17,18] Moreover, sleep-onset problems associated with stress and pre-sleep mentation predominate in athletes, especially before competition.[19] It should be acknowledged that psychological problems, such as depression and anxiety, could increase in athletes undergoing weight loss prior to competition.[20] These mood states have the potential to further exacerbate athletes' precompetitive sleep disturbance.

5. Conclusion and suggested scope

Obsessive use of CSTs may lead to insomnia in athletes through orthosomnia and nomophobia in combination. Nevertheless, studies investigating the prevalence of orthosomnia, nomophobia, mental health symptoms (e.g., anxiety, depression), and insomnia, as well as their associations in athletes, are warranted.

Athletes may suffer from insufficient sleep quantity and poor sleep quality mainly because of precompetitive anxiety and/or stress.[17] However, knowing that their pre-performance night sleep metrics were not optimal may adversely impact their sense of self-efficacy and motivation, resulting in performance decrement. In addition, athletes' pre-performance mental state may be further compromised if their CST includes a feature that proports to measure "readiness" or "recovery" based primarily on sleep measurements.[4] Therefore, mental performance coaches of perfectionist athletes should be aware of those using CSTs in order to provide them cautionary training regarding their accuracy and meaning as well as to provide affected athletes with appropriate pre-performance mental preparation.

Hundreds of millions of people already use CSTs to assess sleep and other biometric data, and this trend is rapidly rising.[10] However, the lack of scientific validation (i.e., reported in peer-reviewed publications) of many CSTs is a significant shortcoming.[1]

To date, individuals using some CSTs are required to provide information about factors that can influence sleep, such as the quantity and the timing of caffeine intake. However, multidisciplinary teams (e.g., sleep and sports scientists, electronic device makers) should work together to develop a new generation of fitness trackers that minimize harmful effects on athletes' sleep and performance. This might be successfully achieved by (a) developing a scientifically validated scale to measure orthosomnia in athletes, a possible cause of sleep disturbances, and (b) incorporating the orthosomnia feature into CST to provide personalized (i.e., based on the athlete's particular characteristics) prevention and intervention programs aimed at improving sleep or counteracting the potentially negative effect of the precompetitive period on sleep metrics.

Authors' contribution

KT and HJ wrote the main manuscript text; all co-authors read and critically revised manuscripts drafts. All authors have read and approved the final version of the manuscript, and agree with the order of presentation of the authors.

Declaration of Competing Interest

The authors declare that they have no competing interests.

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